

CUSTOMER SUPPORT TICKET PRIORITY PREDICTION AND AUTOMATED ASSIGNMENT SYSTEM USING AGENTFORCE

1. INTRODUCTION

1.1 Project Overview

The Customer Support Ticket Priority Prediction and Automated Assignment System using Agentforce is a Salesforce-based solution designed to automate the analysis and prioritization of customer support tickets.

The system uses Salesforce, Salesforce Flow, and Agentforce to analyze the description of a customer's latest support ticket and classify it as High, Medium, or Low priority. Based on the priority, the system can automatically perform appropriate support actions.

For High-priority tickets, the system creates an urgent handling task and assigns the ticket to a senior support agent. Agentforce provides a conversational interface through which users can provide an Account Name and receive the ticket priority and status information.

1.2 Purpose of the Project

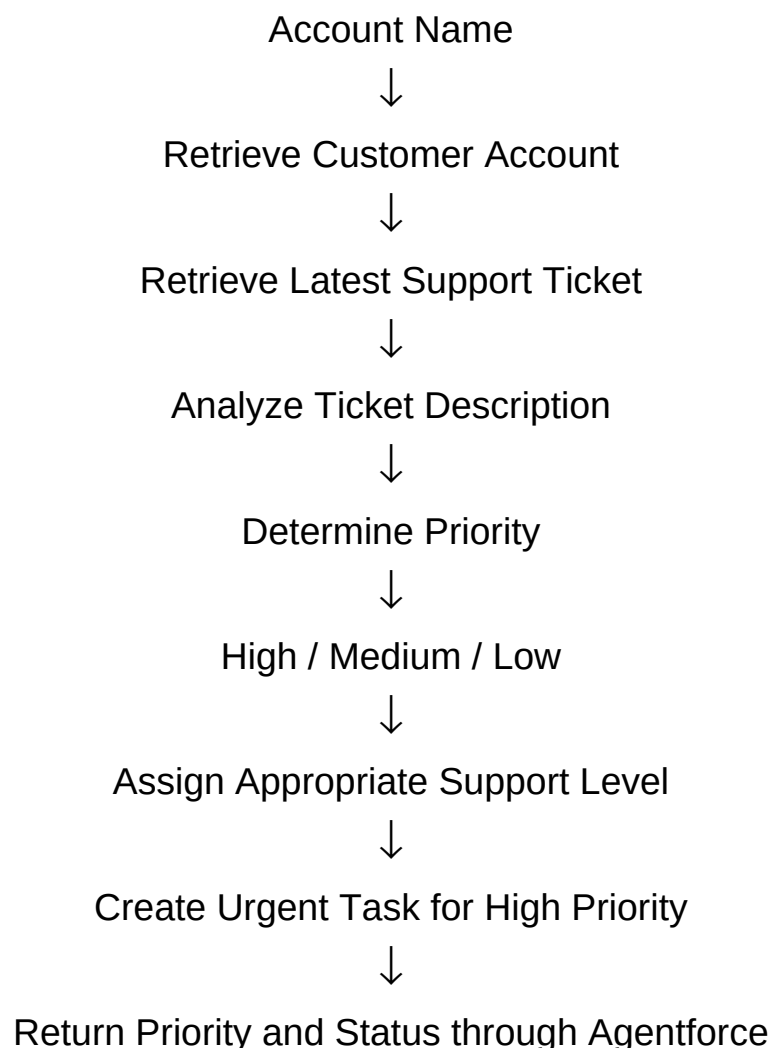
The main purpose of this project is to reduce manual effort in customer support ticket prioritization and assignment. It helps support teams identify critical issues quickly and ensures that high-priority tickets receive

appropriate attention.

The project aims to:

- Automatically classify support tickets as High, Medium, or Low priority.
- Retrieve the latest support ticket associated with a customer account.
- Reduce manual ticket-prioritization effort.
- Automatically create an urgent task for High-priority tickets.
- Assign High-priority tickets to a senior support agent.
- Provide ticket-priority information through Agentforce.

1.3 How the System Works



The solution combines Salesforce data management, Flow automation, and Agentforce to provide an automated approach to customer support ticket prioritization and assignment.

2. IDEATION PHASE

2.1 Problem Statement

The ideation phase identifies the major problems faced by customers, support agents, and support managers while handling customer support tickets. The following problem statements were identified for the proposed system.

PS-1 – Support Agent

I am: Support Agent

I'm trying to: Focus on critical tickets first.

But: Tickets are not always prioritized automatically.

Because: Ticket prioritization can be manual.

Which makes me feel: Frustrated.

PS-2 – Customer

I am: Customer

I'm trying to: Get urgent issues resolved quickly.

But: Critical issues may wait behind lower-priority tickets.

Because: Support teams handle large ticket volumes.

Which makes me feel: Concerned.

PS-3 – Support Manager

I am: Support Manager

I'm trying to: Assign tickets to the right support level.

But: Assignment may require manual decisions.

Because: Ticket urgency is not consistently analyzed.

Which makes me feel: Overwhelmed.

PS-4 – Support Agent

I am: Support Agent

I'm trying to: Identify urgent issues from descriptions.

But: Important keywords can be missed.

Because: Ticket descriptions contain different urgency signals.

Which makes me feel: Uncertain.

PS-5 – Support Manager

I am: Support Manager

I'm trying to: Reduce SLA-related risk.

But: Older unresolved tickets may remain unnoticed.

Because: SLA risk needs additional checking.

Which makes me feel: Concerned.

2.2 Empathy Map Canvas

User Persona: Support Agent

The empathy map helps understand the needs, expectations, difficulties, and benefits experienced by a support agent while handling customer support tickets.

Think & Feel

The support agent wants critical customer issues to be identified quickly and handled according to their urgency.

Hear

The support agent receives customer complaints, issue descriptions, support requests, and information related to ticket status.

See

The support agent works with:

- Customer accounts
- Support tickets
- Issue types
- Ticket descriptions
- Priority levels
- Ticket status

- Assigned agents
- SLA risk information

Say & Do

The support agent:

- Reviews customer support tickets.
- Identifies urgent issues.
- Handles assigned tickets.
- Focuses on high-priority customer problems.
- Works towards timely issue resolution.

Pain

The major problems faced by the support agent include:

- Manual ticket prioritization.
- Difficulty identifying urgency from ticket descriptions.
- Missed critical issues.
- Delays in handling high-priority tickets.
- Additional effort required for ticket assignment.

Gain

The proposed system provides:

- Automatic priority classification.
- Faster identification of critical tickets.
- Clearer ticket assignment.

- Automatic urgent-task creation.
- Reduced manual effort.
- Better support-ticket handling.

2.3 Brainstorming & Idea Prioritization

During the brainstorming phase, different approaches were considered to improve the customer support ticket handling process.

Idea 1 – Manual Ticket Registers

Support agents manually record customer tickets and determine their priority.

Limitation: This requires continuous manual effort and may result in inconsistent prioritization.

Idea 2 – Spreadsheet-Based Ticket Tracking

Customer support tickets can be maintained and monitored using spreadsheets.

Limitation: Spreadsheet-based tracking provides limited automation and may become difficult to manage as ticket volume increases.

Idea 3 – Standalone Support Portal

A separate support portal can be developed to manage customer support requests.

Limitation: Developing and maintaining a separate system requires additional resources and infrastructure.

Idea 4 – Salesforce CRM with Agentforce

Salesforce CRM can be combined with Agentforce and Salesforce Flow to analyze support tickets, determine their priority, automate assignment, and create urgent tasks.

Benefit: It provides CRM data management, automation, and AI-assisted interaction within a single platform.

Selected Idea

Salesforce CRM with Agentforce was selected as the proposed solution.

The selected approach combines:

- Salesforce CRM
- Salesforce Flow automation
- Agentforce
- Automatic priority classification
- Automatic urgent-task creation
- Support-agent assignment

This approach addresses the identified problems by reducing manual ticket prioritization and helping support teams focus on critical customer issues.

FINAL OUTPUT

The final implementation of the Customer Support Ticket Priority Prediction and Automated Assignment System using Agentforce was tested in Salesforce. The following screenshots show the configured Flow, the Agentforce interaction, and the resulting support-ticket details.

1. Salesforce Flow – Priority Analysis and Assignment

The Auto-Launched Flow retrieves the customer account and latest support ticket, analyzes the ticket description, determines the priority, and follows the corresponding High, Medium, or Low path. The High-priority path includes urgent task creation and support-agent assignment.

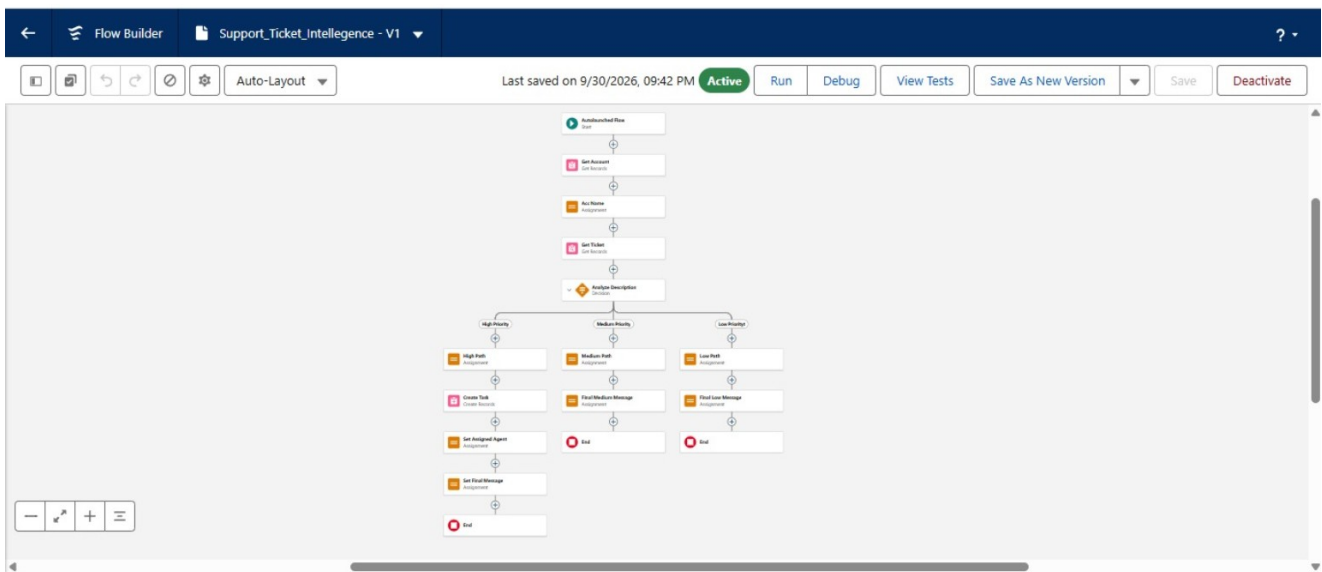


Figure 1. Configured Salesforce Flow for support-ticket priority analysis and automated assignment.

2. Agentforce – Ticket Priority Result

Agentforce was tested by providing the customer account name. The Agentforce panel returned the analyzed ticket priority as High, confirming the conversational output from the configured support-ticket analysis action.

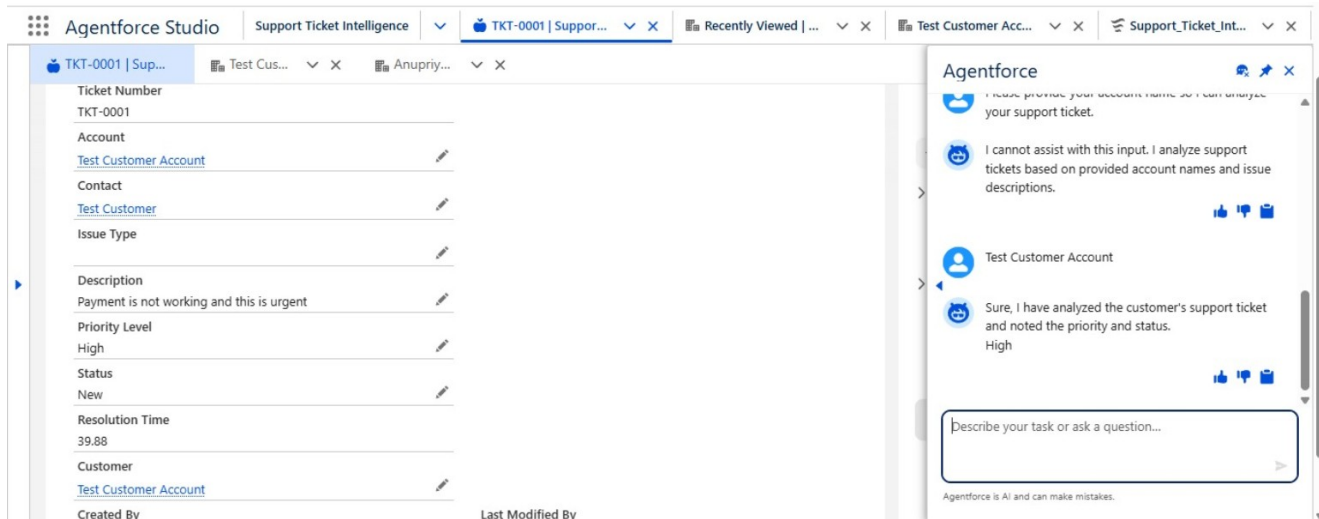


Figure 2. Agentforce interaction showing the analyzed ticket priority as High.

3. Support Ticket Record – Final Result

The support ticket record displays the customer account, contact, ticket description, priority level, status, and other ticket information. The shown record has a High priority level, providing the Salesforce record-level output of the implemented system.

The screenshot displays a Salesforce interface for a Support Ticket Intelligence record. The record ID is TKT-0002. The left sidebar shows a list of related records under the 'Details' tab, including the Support Ticket Intelligence Name, Ticket Number, Account, Contact, Issue Type, Description, Priority Level, Status, and Resolution Time. The main content area shows the ticket details, including the Owner (ANUPRIYA R) and the Activity section. The Activity section shows a list of activities with filters for All time, All activities, and All types. The activity list is currently empty, with a message stating 'No activities to show. Get started by sending an email, scheduling a task, and more.' A 'Show All Activities' button is visible at the bottom of the activity section.

Related	Details
Support Ticket Intelligence Name	TKT-0002
Ticket Number	TKT-0001
Account	Test Customer Account
Contact	Test Customer
Issue Type	
Description	Customer is unable to access the application.
Priority Level	High
Status	New
Resolution Time	25.41

Figure 3. Salesforce Support Ticket Intelligence record showing the final ticket details and High priority.